

Event-Driven Simulation and Modeling of Phase Noise of an RF Oscillator

Robert Bogdan Staszewski, *Member, IEEE*, Chan Fernando, and Poras T. Balsara, *Senior Member, IEEE*

Abstract—A novel simulation technique that uses an event-driven VHDL simulator to model phase noise behavior of an RF oscillator for wireless applications is proposed and demonstrated. The technique is well suited to investigate complex interactions in large system-on-chip systems, where traditional RF and analog simulation tools do not work effectively. The oscillator phase noise characteristic comprising of flat electronic noise, as well as, upconverted thermal and $1/f$ noise regions are described using time-domain equations and simulated as either accumulative or nonaccumulative random perturbations of the fundamental oscillator period. The VHDL simulation environment was selected for its high simulation speed, the direct correlation between the simulated and built circuits and its ability to model mixed-signal systems of high complexity. The presented simulation technique has been successfully applied and validated in a Bluetooth transceiver integrated circuit fabricated in a digital 130-nm process.

Index Terms—Bluetooth, event driven, jitter, modeling, oscillator, phase noise, simulation, system on chip (SoC), VHDL, wireless.

I. INTRODUCTION

ADVANCED CMOS process lithography nowadays allows the creation of fine geometry, but well-controlled, varactors. The switchable capacitance of the finest differential varactor in the 130-nm CMOS process is only of the order of tens of attofarads. A novel *digitally controlled oscillator* (DCO), which deliberately avoids any analog tuning voltage controls, was recently presented in [1]. Fine frequency resolution is achieved through high-speed $\Sigma\Delta$ dithering. This allows for its loop control circuitry to be implemented in a fully digital manner, as proposed in [2]. Other imperfections of analog circuits could be compensated through digital means as described in [3].

Fig. 1 reveals major highlights of a novel digitally intensive architecture of a commercial single-chip Bluetooth radio realized in a 130-nm CMOS process with no analog extensions. The transceiver consists of the first ever reported all-digital RF frequency synthesizer and transmitter section [4], as well as, the first ever reported direct RF sampling discrete-time receiver section [5]. The architecture features integration of RF, analog baseband, digital baseband, memory, power management, and *RF built-in self test* (RFBIST) in a system on chip (SoC) that

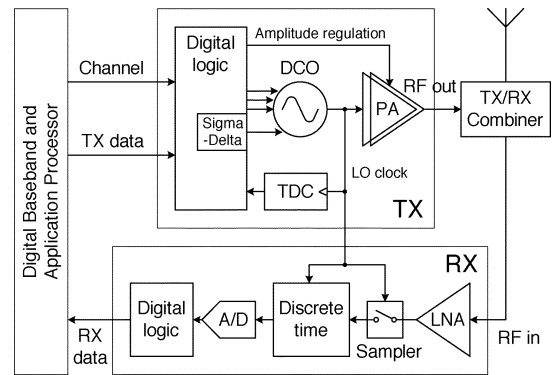


Fig. 1. Single chip Bluetooth radio with an all-digital transmitter [4] and a discrete-time receiver [5].

meets all of the Bluetooth specifications and is amenable to migration to newer deep submicrometer CMOS processes.

With the first demonstration of a fully digital frequency synthesizer and transmitter, and a digitally intensive receiver for RF applications, a need has arisen to model and simulate RF components using the same simulation engine as that used for the digital back-end, which nowadays is likely to contain over a million gates. This way, complex interactions and performance of the entire SoC integrated circuit (IC) could be validated and verified prior to tape out. Fig. 1 offers some examples of these complex interactions.

- 1) Effect of the DCO phase noise on the phase-locked loop (PLL) phase noise performance and generated spurs, especially when the PLL contains a higher-order digital loop filter and operates in fractional- N mode.
- 2) Effect of the DCO frequency resolution on the close-in phase noise of the PLL.
- 3) Effect of the $\Sigma\Delta$ DCO dithering on the far-out phase noise.
- 4) Effect of the DCO varactor mismatches on the modulated spectrum.
- 5) Effect of the DCO phase noise on the degradation of the signal-to-noise ratio in the direct RF sampling receiver.

While SPICE-based simulation tools are extremely useful for small RF circuits containing several components (such as an RF oscillator), their slow simulation performance prevents from investigating larger circuits (such as an RF oscillator with a PLL loop and a transmitter or a receiver). In fact, using the proposed techniques, we were able to determine that the entire radio (with over a million gates) meets the RF Bluetooth specifications prior to tapeout. This level of validation seems to be nowadays a requirement given almost a million dollar price tag for the reticle set in the most advanced deep-submicrometer CMOS process.

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R. B. Staszewski and C. Fernando are with the Wireless Analog Technology Center, Texas Instruments Incorporated, Dallas, TX 75243 USA (e-mail: b-staszewski@ti.com).

P. T. Balsara is with the Center for Integrated Circuits and Systems, University of Texas at Dallas, Richardson, TX 75083 USA.

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There have been a variety of communication channel modeling methods. At the pure system level there are C and MATLAB models, which are highly abstract and with very weak links to the actual hardware. At the other end of the spectrum, a system could be modeled at a very low level entirely in SPICE for analog-intensive systems or in SPICE and Verilog (or VHDL) combination, with a varying degree of link between the two disparate simulation engines. Establishing a link to a nonevent-driven engine, such as SPICE or in analog mixed-signal VHDL (VHDL-AMS), results in a hefty price on simulation performance, thus making it impossible to determine the very basic figure of merit of a communication channel: bit error rate.

This paper describes a behavioral modeling and simulation environment that is based on a standard event-driven single-core simulator, e.g., VHDL. The proposed system is well suited for digitally intensive single-chip integrated solutions with a fair amount of analog circuitry. The main advantage of the single simulation engine at the top level is that it allows seamless integration of all hardware abstraction levels (such as behavioral, RTL, gate level) in a uniform environment. The single most important feature of the standard VHDL hardware description language, which makes it far superior to Verilog for mixed-signal designs, is its support of real or floating-point type signals. Extensive simulation and synthesis support by the standard VHDL language makes it possible for a complex communication system to achieve “build what we simulate, and simulate what we build” goal. Simulator performance, stability, multi-vendor support, mature standard and widespread use are all advantages of this environment.

The organization of this paper is as follows. Basic time-domain equations are derived in Section II. An illustration of phase noise in oscillators is presented in Section III. Section IV discusses event-driven feature of VHDL for modeling and simulation of narrowband RF systems. Time-domain modeling of oscillator phase noise is presented in Section V. An application of the proposed techniques for modeling a DCO is presented in Section VI. Simulated and measured results are shown in Section VII.

II. BASIC TIME-DOMAIN EQUATIONS

Let the nominal frequency of oscillation be f_0 . It is related to the nominal clock period T_0 by its inverse $f_0 = 1/T_0$. If the clock period is shortened by ΔT , the new clock period will be $T = T_0 - \Delta T$. This will result in a higher frequency of oscillation of $f = f_0 + \Delta f$. For $\Delta T/T_0 \ll 1$, the relationship between Δf and ΔT is linear

$$\Delta f \approx f_0 \frac{\Delta T}{T_0} = f_0^2 \Delta T = \frac{\Delta T}{T_0^2}. \quad (1)$$

Equation (1) was used extensively in this design as a conversion formula for system analysis and simulation. The simulation environment is VHDL which, being an event-driven digital simulator, is foreign to the concept of frequency and exclusively operates in the time domain.

Table I relates [based on (1)] the oscillator frequency deviation due to 1 femtosecond of a period deviation for several

TABLE I
TIMING DEVIATION VS. FREQUENCY DEVIATION AT DIFFERENT POINTS IN BLUETOOTH BAND

Period deviation ΔT	Center freq. f_0	Freq. deviation Δf
1 fs	2402 MHz (first channel)	5770 Hz
1 fs	2440 MHz (middle channel)	5953 Hz
1 fs	2480 MHz (last channel)	6159 Hz

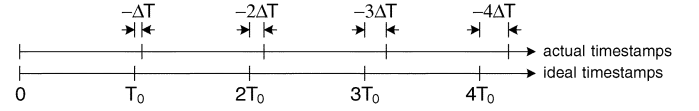


Fig. 2. Development of an accumulative TDEV.

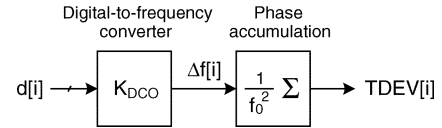


Fig. 3. DCO time-domain model.

frequencies in the Bluetooth band. This makes apparent that a fine timing resolution is required at RF frequencies for time-domain simulation tools. In fact, it was necessary to resort to the finest timing resolution of 1 fs that the VHDL standard provides. From a physical viewpoint, a femtosecond time deviation is quite meaningless for a single observation,¹ and only an averaged value could make sense.

For a time-invariant oscillator with a fixed frequency excursion, ΔT period deviation from T_0 will result in ΔT deviation from ideal timing instances within one oscillator clock period, $2\Delta T$ within two clock periods, etc., as shown in Fig. 2. For varying ΔT values, within i oscillator clock cycles, the accumulated *timing deviation* (TDEV) will reach

$$\text{TDEV}[i] = \sum_{l=1}^i \Delta T[l] = \sum_{l=1}^i \frac{\Delta f[l]}{f_0^2}. \quad (2)$$

Equation (2) states that the TDEV, defined as the difference between actual and ideal timing instances, is an integral of the oscillator frequency deviation. The ΔT direction is selected toward shortening the period such that ΔT and Δf signs agree.

An example of how the time-domain model can be applied to a DCO is presented in Fig. 3. The *oscillator tuning word* (OTW) at the DCO input will change its operating frequency by $\Delta f[i] = d[i] \cdot K_{\text{DCO}}$. On every rising DCO edge event, the DCO event output Δf multiplied by a “constant” $1/f_0^2$ will be accumulated. At the end of i cycles, it will accumulate the TDEV according to (2). The accumulated TDEV is only defined at the end of the DCO clock cycle with each rising clock edge. It should be noted that because the phase is fundamentally a time integral of frequency, the DCO phase accumulation is a pure time development process. The TDEV is a measure of “badness” and signifies the departure from the desired timing instances that has to be corrected by a feedback loop mechanism.

¹One femtosecond is such a small period of time that, during which, the visible light could only travel less than its wavelength.

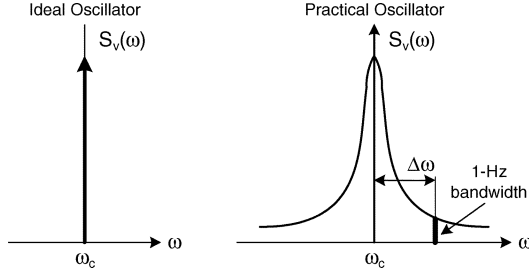
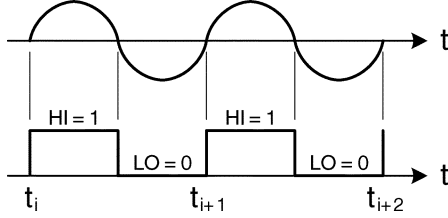
Fig. 4. Voltage output spectrum $S_v(\omega)$ of ideal and practical oscillators.

Fig. 5. Possible outputs of an oscillator: sinusoidal and digital waveforms.

III. PHASE NOISE IN OSCILLATORS

Phase noise is normally characterized in the frequency domain [6], [7]. For an ideal oscillator operating at frequency ω_0 , the voltage output can be expressed as $v(t) = A \cos(\omega_0 t + \phi)$, where A is the amplitude and ϕ is an arbitrary, but fixed, phase reference. The power is concentrated at a single frequency ω_0 . Consequently, its spectrum $S_v(\omega) = (A^2/2)\delta(\omega - \omega_0)$ is the shape of a Dirac impulse as depicted in Fig. 4. In a practical oscillator, however, both the amplitude and phase are time-varying fluctuations and the spectrum will exhibit a “skirt” around the carrier frequency and spread into nearby frequencies. In most cases, the disturbance in the amplitude is negligible or unimportant, since it is customarily removed by a limiter circuit, and, therefore, only the random deviation of the phase is considered

$$v(t) = A \cos(\omega_0 t + \phi(t)) \quad (3)$$

where $\phi(t)$ is a small random excess phase representing variations in the period and is commonly called “phase noise.”

As Fig. 5 shows, the frequency and phase information is preserved in either a continuous-time waveform fit to the ideal sinusoid or in edge transition instances, respectively. A clear advantage of the rectangular digital signal is taken here as more useful for a digital deep-submicrometer CMOS process technology, which enjoys excellent resolution in the time domain. In fact, the significant (either positive or negative) zero-crossing transition timestamps are the only necessary information to represent a narrowband RF signals in GFSK or GMSK communications. This method favors the event-driven feature of VHDL, which spends computational effort only on these timestamp events. In addition, there is no need for post-processing tools to restore the RF waveform. Phase noise and RF spectrum could be obtained exclusively based on the zero-crossing timestamp information. The RF waveform could possibly be restored but it would be disadvantageous from a computational standpoint.

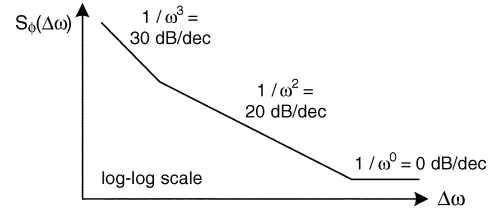


Fig. 6. Phase noise spectrum of an actual oscillator.

For a small value of the phase noise fluctuation $|\phi(t)| \ll 1$ rad, which is quite a good approximation for RF applications, (3) could be simplified to

$$v(t) \approx A \cos \omega_0 t - A \phi(t) \sin \omega_0 t \quad (4)$$

which means that the spectrum of $\phi(t)$ is frequency translated to $\pm\omega_0$.

To quantify this phase noise, we consider a 1-Hz unit bandwidth at an offset of $\Delta\omega$ from the carrier, calculate the noise power in the band, and divide this result by the carrier power [7]. This is the single-sided spectral noise density in dBc/Hz

$$\mathcal{L}\{\Delta\omega\} = 10 \log \left(\frac{\text{noise power in 1-Hz BW at } \omega_0 + \Delta\omega}{\text{carrier power}} \right). \quad (5)$$

The single-sided phase noise is simply one-half of the phase noise spectrum, which contains both upper and lower frequency components

$$\mathcal{L}\{\Delta\omega\} = \frac{S_\phi(\Delta\omega)}{2}. \quad (6)$$

Fig. 6 shows a typical oscillator phase noise spectrum. In this log-log plot, phase noise normalized to dBc/Hz is plotted against the offset frequency $\Delta\omega$ from the carrier ω_0 . The phase noise profile follows the curve shown, where it traverses through $1/\omega^3$, $1/\omega^2$ and $1/\omega^0$ slope regions. The region $1/\omega^2$ is generally referred to as the thermal noise region, since it is caused by white or uncorrelated timing fluctuations in the period of oscillation.

The $1/f$ flicker noise of electronic devices is also substantial for lower offset frequencies and it features a 10 dB/dec slope in a log-log amplitude power spectral density plot. For RF oscillators, (1) describes a linear relationship between the oscillation period excursion and the resulting frequency deviation. Consequently, if the $1/f$ noise affects the oscillation period, it will exhibit itself as a 10 dB/dec slope in the power spectral density of the instantaneous frequency deviation of the oscillator output. On the other hand, the spectrum of the phase noise will exhibit a 30-dB/dec slope. The additional 20-dB/dec attenuation of the phase noise is due to the frequency-to-phase conversion (the conventional $1/s$ operator) in the oscillator during the up-conversion and can be summarized by the following formula:

$$S_\phi(\Delta\omega) = \frac{S_{\Delta\omega}(\Delta\omega)}{\omega^2} \quad (7)$$

where $S_{\Delta\omega}(\Delta\omega)$ is the power spectral density of the instantaneous frequency deviation. Finally, the $1/\omega^0$ region is the

thermal electronic noise added to the clock outside of the oscillator, such as in an output buffer, and which does not affect the oscillator time base.

IV. BEHAVIORAL MODELING AND SIMULATION IN VHDL

VHDL is based on an event-driven simulation engine. The simulator proceeds to the timestamp of the next event if all the activities associated with the current timestamp are exhausted. It is a very efficient method since the simulation activity (hence, computation time) is only spent on a per-needed basis. This is in sharp contrast with some other system-level simulators which are based on an oversampled clock, such as Simulink or SPW. In that environment, the simulation engine has to transverse all the equally spaced timestamps that sufficiently oversample the signals.

Operating in an oversampled domain is less problematic with baseband signals and systems or in an environment with a single clock. Even two clock domains are not an issue if their frequency ratio is a small integer. In that case, the higher frequency clock is the common denominator. The operation becomes quite unwieldy, however, if the clocks are not easily related such that their common denominator clock is at a very high frequency.

Another environment exposing inefficiency of the oversampled domain simulator is a narrowband RF system. Oversampling a Bluetooth RF waveform of the 2.4-GHz carrier generates an excessive amount of activity in light of the fact that the information is contained only in the 1 Mbps symbols. Considering eight samples per sinusoidal RF cycle, one 1 μ s symbol would contain as many as 19 200 samples! We have chosen a very efficient RF wave representation method in which only positive zero-crossing timestamps (t_i, t_{i+1}, t_{i+2} in Fig. 5) are used.

V. TIME-DOMAIN MODELING OF PHASE NOISE

Phase noise of the oscillator is modeled using jitter and wander constructs. The flat electronic noise of the $1/\omega^0$ region in Fig. 6 is modeled as a nonaccumulative jitter. The $1/\omega^2$ region of the upconverted thermal noise, on the other hand, is modeled as an accumulative wander. The $1/\omega^3$ region in Fig. 6 of the upconverted $1/f$ noise is modeled as a weighted sum of weakly overlapping wander contributions.

A. Modeling of the Oscillator Jitter

Fig. 7 illustrates the modeling principle for the timing jitter. The $1T_0, 2T_0, 3T_0, 4T_0$ timestamps are the ideal equidistant rising-edge events of an oscillator operating at frequency $f_0 = 1/T_0$. The ideal oscillator output might pass through a physical buffer that adds random fluctuations to its delay. The actual timestamps of the physical buffer output could be described mathematically as an additive random error at each occurrence of the ideal timestamp. These timing errors do not influence one another. If the random error is due to the thermal electronic noise, then the edge TDEVs are said to be *independent and identically distributed* (iid) and *are usually modeled as an additive white Gaussian noise* (AWGN). Fig. 7 shows the error probability curve from each ideal timestamp.

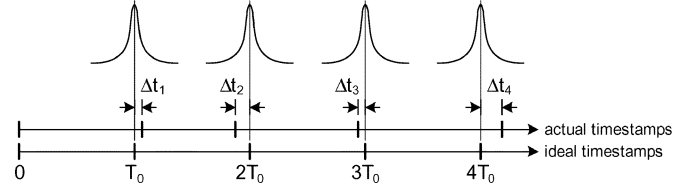


Fig. 7. Development of nonaccumulative TDEV.

The timing deviation TDEV_j in the jitter case is the difference between the actual timestamps $t_j[i]$

$$t_j[i] = i \cdot T_0 + \Delta t[i] \quad (8)$$

and the ideal timestamps at $i \cdot T_0$:

$$\text{TDEV}_j[i] = \Delta t[i]. \quad (9)$$

The period deviation $p_1[i] = t_j[i] - t_j[i-1]$ is statistically twice the variance of the timestamp deviation. This is due to the fact that an instantaneous period is perturbed from both sides and makes the neighboring errors not entirely independent. The period deviations could be modeled by passing the timestamp deviations through a 2-tap finite-impulse response (FIR) filter with $\{1, 1\}$ coefficients.

The relationship between the time and frequency domains of the jitter could be obtained as follows. The noise floor $\mathcal{L}$ is a single-sided spectral density. It needs to be multiplied by two in order to arrive at the S_ϕ [rad^2/Hz] double-sided spectral density in (6). Since the flat spectrum of the jitter in the discrete-time model extends to the Nyquist frequency, S_ϕ is multiplied by $f_0/2$ to arrive at the total power [rad^2]. The rms jitter [rad] is its square root value. The radian-unit quantity is converted to the timestamp deviation [sec] by multiplying it by the normalizing factor of $T_0/2\pi$. The above results in the following:

$$\sigma_{\Delta t} = \frac{T_0}{2\pi} \cdot \sqrt{\mathcal{L} \cdot f_0}. \quad (10)$$

For Bluetooth, substituting $f_0 = 2.4$ GHz, $T_0 = 417$ ps, and $\mathcal{L} = 10^{-150} \text{ dB}/10 = 1 \cdot 10^{-15} \text{ rad}^2/\text{Hz}$, it amounts to $\sigma_{\Delta t} = 103$ fs. Per (1), this figure corresponds to 613-kHz frequency deviation. It reveals that one standard deviation of the jitter is several times higher than the full symbol frequency deviation of 160 kHz. In the case of GSM, the noise floor has to be at least below $-164 \text{ dBc}/\text{Hz}$. Substituting, $f_0 = 900$ MHz, $T_0 = 1111$ ps, and $\mathcal{L} = 10^{-164} \text{ dB}/10 = 4 \cdot 10^{-17} \text{ rad}^2/\text{Hz}$, the rms jitter is $\sigma_{\Delta t} = 33$ fs.

B. Modeling of the Oscillator Wander

Fig. 8 illustrates the modeling principle for the timing wander, which is also called an accumulative jitter. This could be visualized by a physical oscillator of nominal period T_0 , whose actual period varies slightly from one cycle to the next, due to, for example, thermal noise effects that are internal to the oscillator. In contrast to the jitter case, each transition timestamp does depend here on all previous period deviations. This simply acknowledges the fact that the ideal timestamps merely exist in theory and the only memory of a given transition is the previous transition timestamp. This behavior is modeled as a random walk.

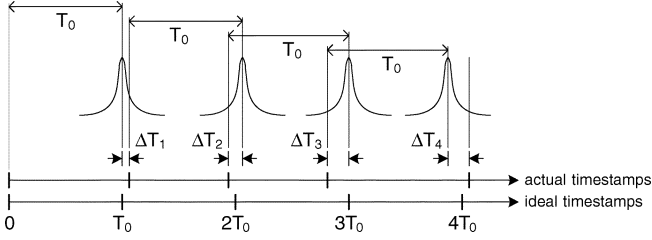


Fig. 8. Development of an accumulative TDEV.

The timing deviation TDEV_w in the wander case is the difference between the actual timestamps $t_w[i]$

$$t_w[i] = i \cdot T_0 + \sum_{l=1}^i \Delta T[l] \quad (11)$$

and the ideal timestamps at $i \cdot T_0$

$$\text{TDEV}_w[i] = \sum_{l=1}^i \Delta T[l]. \quad (12)$$

It should be noted that other references use different terms for “wander.” For example, [8] uses the term “absolute jitter.” Unlike in the nonaccumulative case, the period deviation here is equivalent to the timestamp deviation.

Relationship between the wander component in the time and frequency domains is obtained (based on [9]) as

$$\sigma_{\Delta T} = T_0 \cdot \Delta\omega \sqrt{\frac{\mathcal{L}\{\Delta\omega\}}{2\pi\omega_0}} = \frac{\Delta f}{f_0} \cdot \sqrt{T_0} \cdot \sqrt{\mathcal{L}\{\Delta\omega\}}. \quad (13)$$

This equation is also confirmed by [10]. For Bluetooth, substituting $f_0 = 2.4$ GHz, $T_0 = 417$ ps, $\Delta f = 500$ kHz, $\mathcal{L}\{\Delta f\} = 10^{-112}$ dB/10 = $6.3 \cdot 10^{-12}$ rad²/Hz, it results in $\sigma_{\Delta T} = 11$ fs. For GSM, substituting, $f_0 = 900$ MHz, $T_0 = 1111$ ps, and $\mathcal{L} = 10^{-164}$ dB/10 = $4 \cdot 10^{-17}$ rad²/Hz at $\Delta f = 20$ MHz offset, the rms wander is $\sigma_{\Delta t} = 5$ fs.

C. Modeling of the Oscillator Flicker (1/f) Noise

In previous work, $1/f$ noise has been modeled in the time domain by means of FIR and IIR filters [11]. Filter coefficients are derived depending on the type of noise that is modeled. The main disadvantage there is that for high filter sampling rates, if the $1/f$ noise has to be described over several decades, then the required number of filter coefficients becomes very large. For example, for a filter operating at 1 GHz to be able to describe the $1/f$ noise to 1 kHz would require 100 000 filter coefficients.

In this work, the $1/f$ noise is constructed by passing white noise through several first-order low-pass filters. Each filter shapes different regions of the noise spectrum to yield a composite output having the desired $1/f$ response of slope

$$\text{slope} = \frac{A_{\text{dB}}}{r} \quad (14)$$

equal to 10 dB/decade or 3 dB/octave. The $1/f$ noise is then upconverted by the oscillator as per (7), which results in the final 30 dB/dec slope of the phase noise.

Fig. 9 illustrates the modeling principle. The frequency over which the $1/f$ noise is modeled is divided into several regions.

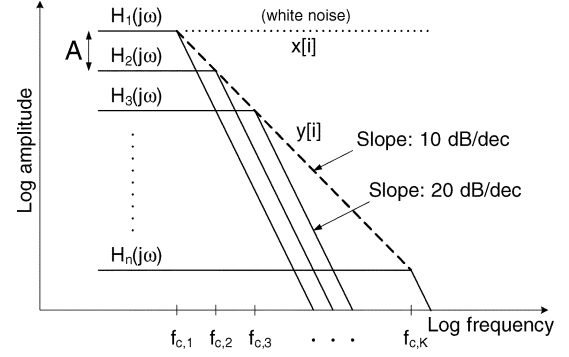
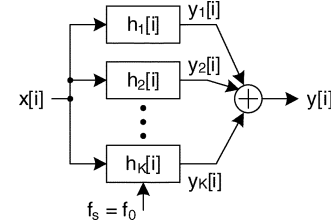


Fig. 9. Composite response of single-pole lowpass filters.

Fig. 10. Time-domain model for the $1/f$ noise generation.

The frequency boundaries between regions serve as the corner frequency of the filters used in the noise shaping process. Ratio of the dc gains between the successive filters

$$A = 10^{\frac{A_{\text{dB}}}{20}} \quad (15)$$

is selected such that the ratio of the corresponding neighboring corner frequencies

$$r = \frac{f_{c,k+1}}{f_{c,k}} \quad (16)$$

satisfies (14). As seen from Fig. 9, the composite response of the filters yields a $1/f$ noise characteristic having the desired slope of 10 dB/decade.

Fig. 10 illustrates a time-domain model for generating the $1/f$ noise. White Gaussian unit-variance noise source $x[i]$ is input to each filter. The outputs of all the filters are summed to obtain the shaped $1/f$ noise response over the frequency interval ranging from $f_{c,1}$ to $f_{c,K}$. Each filter, $h_k[i]$, is modeled as a first-order infinite-impulse response (IIR) filter described by the difference equation

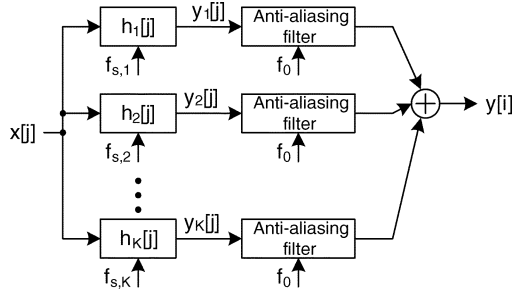
$$y_k[i] = (1 - a_k)y_k[i - 1] + a_k A^{-(k-1)} x[i] \quad (17)$$

where k is the filter index, $k = 1, 2, \dots, K$, A is the linear value of the attenuation factor given in (15), and a_k is defined by

$$a_k = 2\pi \frac{f_{c,k}}{f_s} \quad (18)$$

where $f_{c,k}$ is the corner frequency of the filter k and f_s is the common sampling rate.

If the shaping filters are implemented using a multirate approach and the corner frequencies are scaled with the multirate frequencies, then the filters will have the same feedback coefficients. The multirate approach both simplifies the design of

Fig. 11. Multirate $1/f$ noise configuration.

these filters and reduces the computational complexity. Consider a structure illustrated in Fig. 11. Each of the filters is clocked by a down-divided version of the oscillator frequency, f_0 , such that

$$f_{s,k} = \frac{f_0}{M \cdot \prod_{l=0}^{K-k} r^l} \quad (19)$$

where K is the total number of filter segments. The last filter section has the highest sampling frequency of f_0/M . Each of the subsequent sections have a sampling frequency equal to that of the following section divided by A , such that

$$f_{s,k} = \frac{f_{s,k+1}}{r}. \quad (20)$$

Aliases created due to the use of the lower sampling rates are removed by the anti-aliasing filters located at the outputs of each of the filters. The anti-aliasing filters are clocked at the RF output frequency f_0 and include the decimation scaling gain of $\sqrt{f_{s,k}/f_0}$.

If the corner frequencies are also related by the same sampling ratio so that

$$f_{c,k} = \frac{f_{c,k+1}}{r} \quad (21)$$

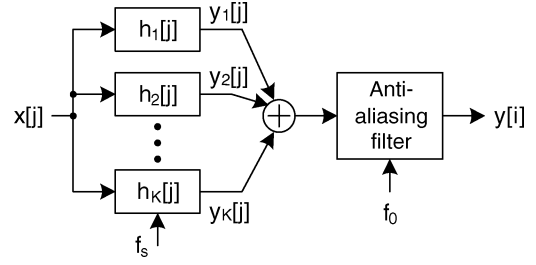
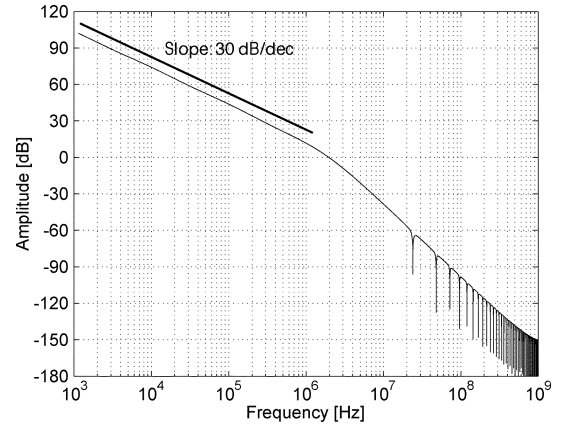
then, all the filters will have the same feedback coefficients. For a first-order IIR filter, the magnitude of the feedback coefficient can be given by

$$a_k = 2\pi \frac{f_{c,k}}{f_{s,k}}. \quad (22)$$

This formula clearly illustrates that if f_s and f_c scale by the same ratio, the filter coefficients remain constant.

A preferred implementation that does not require as many clock domains as in the previous scheme is illustrated in Fig. 12. In this configuration, all the $1/f$ noise shaping filters are driven at the same low clocking frequency f_s and the aliased images are removed by a an anti-aliasing filter located at the cumulative output, which is clocked at the RF clock rate f_0 . The number of antialiasing filters stages used in an implementation depends on the amount of filtering required in the aliased band.

Fig. 13 illustrates the spectral response at the DCO output for an implementation in which the $1/f$ filters have cutoff frequencies at $f_{c,1} = 100$ Hz, $f_{c,2} = 1$ kHz, $f_{c,3} = 10$ kHz,

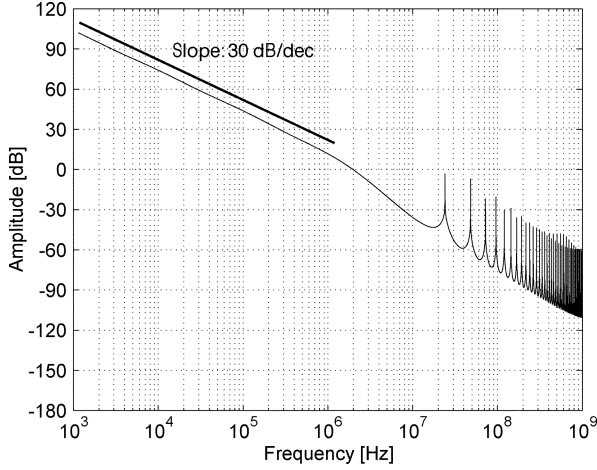
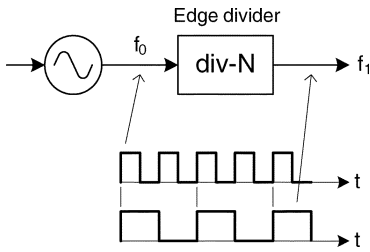
Fig. 12. Multirate $1/f$ noise configuration with single clock domain for all noise shaping filters.Fig. 13. Multirate $1/f$ noise response with sinc filtering.

$f_{c,4} = 100$ kHz and $f_{c,5} = 1$ MHz. This implementation runs at an RF clock of $f_0 = 2400$ MHz and the $1/f$ filters are clocked at $f_s = 24$ MHz. An anti-aliasing filter located at the output of the summer is clocked at the RF clock rate and has a cutoff frequency of 2 MHz. The nulls at multiples of the $1/f$ filter sampling clock arise due to the sinc response that is created by the inherent sample and hold characteristic introduced by the use of multiple clock frequencies.

The sinc response not only provides an extra 20 dB/decade roll-off for frequencies above the $1/f$ filter clocking rate, but also nulls out the aliased frequency components that are present at multiples of this frequency. Fig. 14 illustrates the filtered response without the sinc filtering. As can be seen in this figure in the absence of the sinc filter aliased spurs appear at multiples of the $1/f$ clocking rate.

It should be emphasized that the construction method of the oscillator $1/f$ noise does not fundamentally differ from the wander method. In the former case, the appropriately filtered Gaussian noise $y[i]$ affects the oscillating frequency deviation $\Delta f[i]$. In the latter case, the filtering is not applied and the white Gaussian noise $x[i]$ is used instead, as illustrated in Fig. 9. In both cases, the frequency-to-phase conversion of (7) gives the additional 20-dB/dec slope.

Equation (13) relates the $1/\omega^2$ phase noise $\mathcal{L}\{\Delta\omega\}$ to the timing wander $\sigma_{\Delta T}$. In case of the shaped $1/f$ noise, the phase noise at the corner frequencies is 3 dB less, due to the first-order filtering, than that given by the above equation. For the lowest corner frequency point on the $1/f$ noise curve in Fig. 9, which

Fig. 14. Multirate $1/f$ noise response without sinc filtering.Fig. 15. DCO with an edge divider. In this timing example, $N = 2$.

corresponds to $f_{c,1}$, we calculate the standard deviation using the following:

$$\sigma_{\Delta T, \frac{1}{f}} = \frac{\Delta f_{c,1}}{f_0} \cdot \sqrt{T_0} \cdot \sqrt{2\mathcal{L}\{\Delta\omega_{c,1}\}}. \quad (23)$$

The factor of $\sqrt{2}$ has been included to account for the 3-dB reduction of the phase noise. However, due to the correlative nature of the noise shaping filter outputs, because of their common input, a further correction has to be made to $\mathcal{L}\{\Delta\omega_{c,1}\}$. For the case covered in this paper of $r = 10$, we need to additionally subtract 5.5 dB. If each filter is fed from a separate and uncorrelated noise source, the additional correction would only be -1.5 dB.

D. Clock Edge Divider Effects

RF oscillator is often followed by a clock edge divider, as shown in Fig. 15. This is motivated by the following reasons.

- 1) Quadrature-based receiver architecture requires a precise generation of four 90° spaced clocks. The most straightforward method of generating these clocks is to run the oscillator at double the frequency and use a quadrature divider.
- 2) Decrease coupling of the strong RF power amplifier output back into the oscillator. In this case only the second harmonic of the power amplifier (PA) output, which is much weaker, can affect the oscillator.
- 3) The most optimal quality factor Q of the RF inductor might happen to lie far above the operational RF band.

The analysis below examines how the addition of the clock edge divider affects the phase noise modeling of the oscillator. In case of the flat noise, the jitter contributions are uncorrelated. The same value of the rms jitter is to be used for the oscillator core operating at double the frequency. Since the jitter contributions are uncorrelated, it is OK to throw away every other edge. The rms jitter value in time units does not change in case of clock edge divisions. Another way of looking into this: aliasing by two doubles the noise spectral density in rad^2/Hz units, and, since the period is now half, the noise value in time units does not change. Hence, for the jitter case

$$\sigma_{\Delta t,0} = \sigma_{\Delta t,1}. \quad (24)$$

The equivalent rms wander σ_0 at the oscillator core is lower by $\sqrt{N}$ from the wander σ_1 at the output of the edge divider, where N is the division ratio, according to the law of power addition of independent and identically distributed random variables. Hence, for the wander case

$$\sigma_{\Delta T,0} = \frac{1}{\sqrt{N}} \cdot \sigma_{\Delta T,1}. \quad (25)$$

Due to its linear superposition of the creation technique, the $1/f$ noise follows the same principles as the wander case.

E. Event-Driven Model Realization of the Oscillator

For the event-driven oscillator implementation in VHDL, (8) is rewritten here as

$$t_j[i] = \sum_{l=1}^i (T_0 + \Delta t[l] - \Delta t[l-1]) \quad (26)$$

with $\Delta t[0] = 0$, whereas (11) is rewritten here as

$$t_w[i] = \sum_{l=1}^i (T_0 + \Delta T[l]) \quad (27)$$

where $\Delta t[i]$ and $\Delta T[i]$ are Gaussian-distributed random variables with $\sigma_{\Delta t}$ (10) and $\sigma_{\Delta T}$ (13) standard deviations, respectively. The nonaccumulative and accumulative nature of the noise is readily seen from the above two equations. The jitter and wander (including the $1/f$ noise) contributions, $\Delta t[i]$ and $\Delta T[i]$, are normally combined into one equation

$$t_{jw}[i] = \sum_{l=1}^i (T_0 + \Delta t[l] - \Delta t[l-1] + \Delta T[l]). \quad (28)$$

VI. EXAMPLE VHDL MODEL OF DCO

As a way of an example, we apply the proposed event-driven modeling techniques to an RF DCO that was recently disclosed in [1]. According to Fig. 16, the DCO contains four capacitor banks that are used with the three operational modes: PVT,

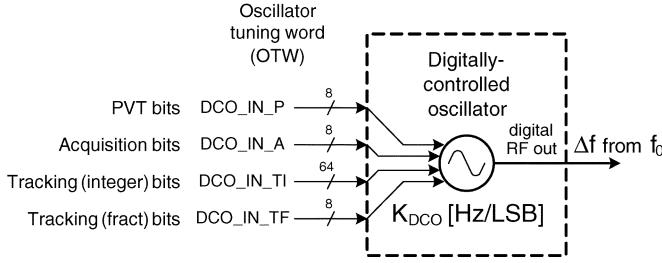


Fig. 16. DCO as an ASIC cell with digital I/Os.

TABLE II
FREQUENCY RESOLUTION AND CORRESPONDING DCO PERIOD DEVIATION FOR
DCO MODES AT THE MIDDLE OF THE BLUETOOTH BAND, $f_0 = 2440$ MHz

Mode	Frequency resolution	Equivalent time resolution
PVT	2326 kHz	390.7 fs
Acquisition	461 kHz	77.43 fs
Tracking	23 kHz	3.863 fs

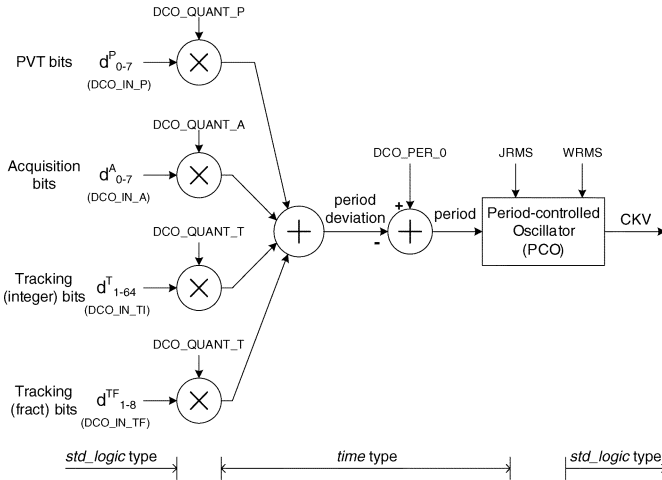


Fig. 17. DCO time-domain model in VHDL.

acquisition and tracking (both integer and fractional tracking banks are used in the tracking mode).

Table II relates the DCO frequency resolution of each capacitor bank to the corresponding DCO period deviation at the middle of the Bluetooth band.

A diagram illustrating a VHDL model of the DCO is shown in Fig. 17. Referring to Fig. 16, DCO_IN_P, DCO_IN_A, DCO_IN_TI and DCO_IN_TF are the digital *std_logic_vector* inputs deviating the DCO oscillating frequency by controlling the *LC* tank capacitance of the PVT, acquisition, tracking-integer and tracking-fractional varactor banks, respectively. Signed-number integer representations of these inputs are multiplied by their respective unit time deviations of the “natural” period: DCO_QUANT_P, DCO_QUANT_A and DCO_QUANT_T. They are VHDL generics rounded off to the closest femtosecond and whose calculated values were shown in Table II for the middle of the Bluetooth band. Their outputs are then summed to create a composite period deviation signal of the VHDL time-type. This signal is then subtracted from the “natural” or center oscillating period DCO_PER_0 since the period deviation is

of opposite sign to the frequency deviation. This time-type signal controls the instantaneous period of the DCO oscillation through a *period-controlled oscillator (PCO)*, whose VHDL code fragment is listed below.

```

1: entity src_pco is
2:   generic (
3:     WANDER_rms: time;
4:     JITTER_rms: time;
5:     SEED: integer
6:   );
7:   port (
8:     period0: in time;
9:     en: in std_logic;
10:    clk: out std_logic
11:  );
12: end entity;
13:
14: architecture behav of src_pco is
15:   signal smp: bit:= '0';
16: begin
17:   process (smp) is
18:     variable initial: boolean:=ture;
19:     -- instantaneous jitter value
20:     variable jitter: time:= 0 ns;
21:     variable jitter_prev: time:= 0 ns;
22:     -- instantaneous wander value
23:     variable wander: time:= 0 ns;
24:     -- the current clock period
25:     variable period: time:= 0 ns;
26:     variable s1: integer:=SEED;
27:     variable randvar: real;
28:   begin
29:     if not initial then
30:       -- adjust the next period
31:       period:=period0;
32:       -- add Gaussian jitter
33:       sub_randn(randvar);
34:       jitter:=randvar}*JITTER_rms;
35:       period:=period + jitter
36:       -jitter_prev;
37:       jitter_prev:=jitter;
38:       -- add Gaussian wander
39:       sub_randn(randvar);
40:       wander:=randvar*WANDER_rms;
41:       period:=period + wander;
42:       -- clock with 50% duty cycle
43:       clk<='1', '0' after period/2;
44:       smp<=not smp after period;
45:     else
46:       period:=period 0;
47:       clk<='0';
48:       -- first transition
49:       smp<='1';
50:       initial:=false;
51:     end if;
52:   end process;
53: end architecture;

```


Input port “period0” of type `time` (line 8) controls the oscillator period during the next cycle. The “smp” internal signal is used to control event activity and to establish the next timestamp (line 43). It is also used to schedule rise and fall times of the “clk” clock (line 42). With each timestamp, jitter and wander values are added at lines 33–36 and 38–40, respectively.

Function calls on lines 33 and 38 use a random number generator with Gaussian (normal) distribution that is based on the Box-Muller method [12]. It uses `uniform()` procedure call of the IEEE `math_real` VHDL package.

VII. SIMULATED AND EXPERIMENTAL RESULTS

The timestamp formulas, (26) and (27), are first validated through simulation. The analysis process runs as follows. The timestamps $t[i]$, as determined in real-time by the “now” VHDL call, are written into a file during the VHDL simulation. In the MATLAB post-processing phase, the timestamps are read and the average period $T_0 = (t[L] - t[1]) / (L - 1)$, where $L = \text{length}(t)$, is calculated. TDEV is obtained as

$$\text{TDEV}[i] = t[i] - i \cdot T_0. \quad (29)$$

Relationship between TDEV and phase noise ϕ is straightforward: $\phi[i] = 2\pi(\text{TDEV}[i]/T_0)$. The final single-sided phase noise spectrum $\mathcal{L}\{\Delta\omega\}$ is obtained as one-half of the power-spectral density of ϕ .

To further validate and show effectiveness of the proposed methodology, a 2.4-GHz DCO and its digital control circuitry have been modeled and fabricated as part of a commercial Bluetooth radio in a digital 130-nm CMOS process.

Fig. 18 (top) shows a simulated synthesizer phase noise spectrum with the DCO oscillator operating in a type-I *all-digital* PLL (ADPLL) with a loop bandwidth of 8 kHz, as described in [2] and [4]. It concentrates on the upconverted thermal and $1/f$ regions (see Fig. 6). The frequency components within the loop filter undergo the 20 dB/dec attenuation. The phase noise readout of -112 dBc/Hz at 500-kHz offset confirms (13). The simulated phase noise well matches the measured results. The electronic noise floor of -150 dBc/Hz (not shown) at high frequencies confirms (10).

Fig. 19 shows the composite trajectory plot of the instantaneous frequency deviation while illustrating operation of various DCO modes. The x axis is the time evolution in DCO clock units (about 417 ps). The y axis is the frequency deviation from an initial value of 2402 MHz (channel 0) expressed in femtosecond time units, where $\Delta T = 1$ fs corresponds to $\Delta f = 5.77$ kHz (see Table I).

The initial starting point is the center frequency set to channel zero. At power-up, a “cold start” to channel four at 4 MHz away is initiated. The ADPLL operates first in the PVT mode by enabling the PVT oscillator controller. This controller makes very coarse (2.3 MHz) adjustments to the frequency. Next, the output of the PVT controller is put on hold and the acquisition oscillator controller is enabled. The acquisition controller quickly brings the frequency near the selected channel in 461-kHz steps. After the acquisition of the selected channel is complete, the output of the controller is put on hold

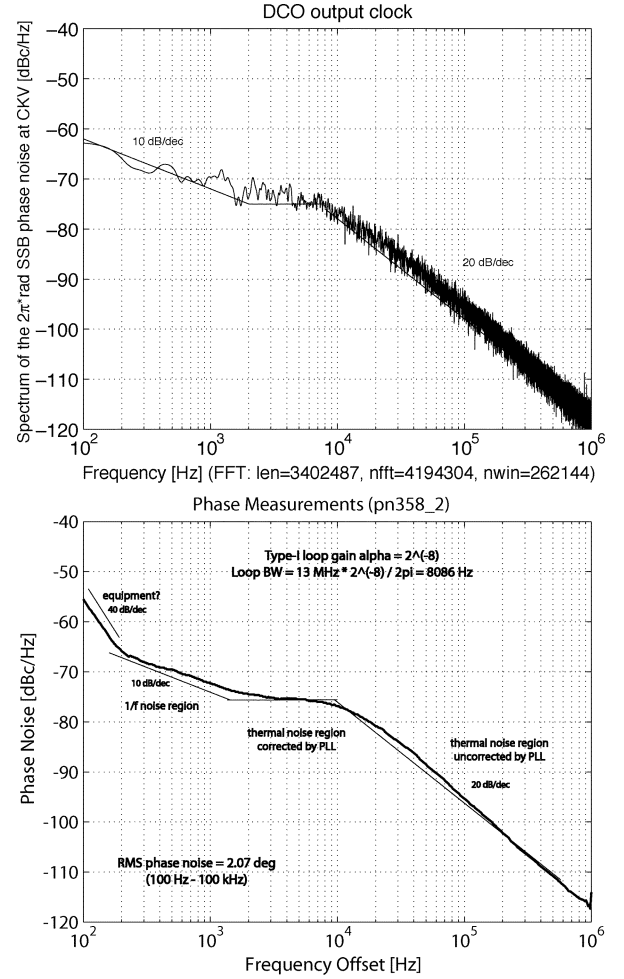


Fig. 18. Synthesizer phase noise with the presented DCO: (top) simulated; (bottom) measured using HP8563E spectrum analyzer with HP85671A phase noise utility.

and the integer and fractional tracking oscillator controller are enabled. The finest selection of the requested channel can only be accomplished using the tracking bank varactors with all the resolution enhancement techniques possible for this capacitor bank [2]. The dynamic range of this mode has to cover the frequency resolution grid of the preceding acquisition mode. In the fast tracking mode, the frequency steps are the finest (less than 1 kHz) but the loop bandwidth could be as fast as in the acquisition mode. The tracking mode featuring the narrow loop bandwidth completes the channel acquisition and frequency locking. The locking process takes altogether $15 \mu\text{s}$ with the reference frequency of 13 MHz (about 36000 DCO cycles or 196 FREF cycles). Upon reaching the steady state of the acquisition, the GFSK data modulation takes place.

VIII. COMPARISON WITH OTHER SIMULATORS

Key advantage of the proposed method is performance. The stand-alone DCO executes $15 \mu\text{s}$ worth of simulation per second on a Sun-Blade-1000 UltraSparc-III workstation. The entire PLL-based transmitter executes $1 \mu\text{s}$ per second of simulation time, so it takes about 10 minutes to simulate the full Bluetooth packed in order to start addressing the complex transmitter and receiver interactions enumerated in Introduction.

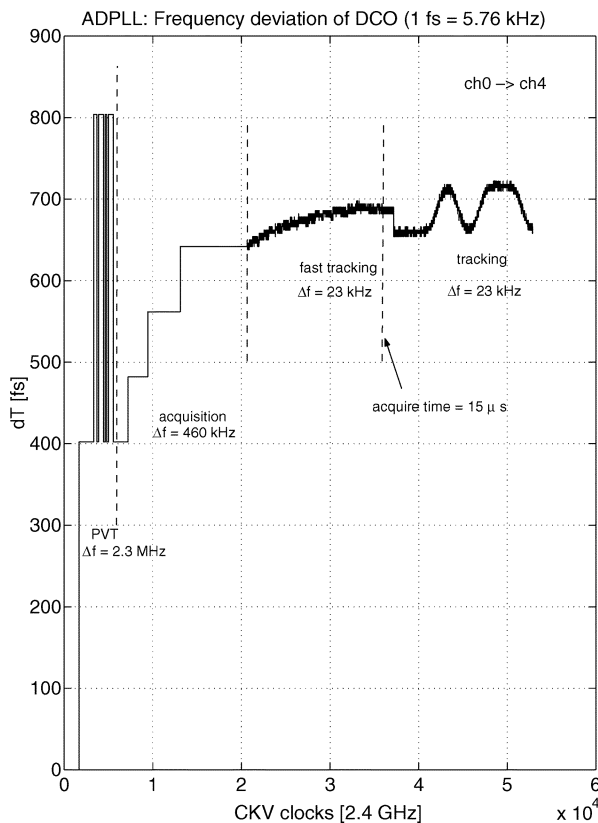


Fig. 19. Simulation plot of the transmit modulation at @2.4 GHz RF output; y-axis: Δf in femtosecond time units (1 fs = 5.76 kHz); x-axis: time in 417 ps RF clock periods.

The SPICE-like alternatives are not practical for higher-level simulations, which usually would take days for 1 μ s of simulated time for a DCO, but would not even converge for the complete ADPLL. The VHDL-AMS alternative as a system simulator has also been evaluated but found to be similarly slow for this SoC.

Apart from the sheer performance issues, the SPICE-like engines are not capable to simulate certain DCO effects, such as the impact of the DCO varactor mismatches on the modulated spectrum. Referring to Tables I and II, a DCO with a representative varactor mismatch of 500 Hz at 2.4 GHz output would require a timing resolution of better than 100 attoseconds, which is clearly not practical with SPICE-like simulators.

IX. CONCLUSION

We have proposed and demonstrated a simulation and modeling methodology of an RF oscillator. The complete phase noise characterization of the oscillator is modeled using time-domain event-driven constructs. A novel method of modeling the upconverted $1/f$ noise is presented. The shown examples apply to a DCO, even though the modeling techniques are readily extendable to a general class of oscillators. The simulator proposed is an event-driven VHDL engine. Sharing the same simulation engine for an RF oscillator with the digital back-end enables to seamlessly employ fully digital frequency synthesizers using sophisticated DSP algorithms, realized in the most advanced deep-submicrometer digital CMOS processes. As part of validation of this methodology, a 2.4-GHz

DCO and its PLL circuitry have been fabricated as part of a Bluetooth radio in a digital 130-nm CMOS process. This paper demonstrates feasibility and attractiveness of employing the event-driven simulation methodology for RF oscillators within large SoC designs.

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Robert Bogdan Staszewski (M'94) received the B.S.E.E. (*summa cum laude*), M.S.E.E. and Ph.D. degrees from the University of Texas at Dallas in 1991, 1992 and 2002, respectively.

From 1991 to 1995, he was with Alcatel Network Systems, Richardson, TX, working on sonnet cross-connect systems for fiber optics communications. He joined Texas Instruments, Dallas, TX, in 1995, where he is currently a Distinguished Member of Technical Staff. Between 1995 and 1999, he has been engaged in advanced CMOS read channel development for hard disk drives. In 1999 he co-started a Digital Radio Processor (DRP) Group within Texas Instruments with a mission to invent new digitally intensive approaches to traditional RF functions for integrated radios in deep-submicrometer CMOS processes. He is currently a design team leader for transmitters and frequency synthesizers in mobile wireless terminals. His research interests include deep-submicrometer CMOS architectures and circuits for frequency synthesizers, transmitters and receivers.



Chan Fernando received the B. Eng., M. Eng., and Ph.D. degrees from Carleton University, Ottawa, ON, Canada, in 1987, 1989 and 1995, respectively.

From 1989 to 1991, he was with Canadian Astronautics Limited, Ottawa, ON, Canada, where he worked on the board level design of circuits to control RF transceivers for imaging radar applications. From 1995 to 1998, he was with Nortel Networks, Ottawa, ON, Canada where he worked on design and verification of wireless local loop terminals and PCS basestations. He joined Texas Instruments Inc,

Dallas, TX, in 1998, where he is involved with the design of RF transceivers for WCDMA, Bluetooth, and GSM standards.



Poras T. Balsara (S'83–M'85–SM'95) received the L.E.E. diploma in electronics from The Victoria Jubilee Technical Institute, Mumbai, India, in 1980, the B.E. (electrical) degree from The University of Bombay, Mumbai, India, in 1983, the M.S. and Ph.D. degrees in computer science from the Penn State University, Philadelphia, in 1985 and 1989, respectively.

Currently, he is a professor of electrical engineering with the Erik Jonsson School of Engineering and Computer Science, The University of Texas at

Dallas. His research interests include, very large-scale integration design, design of energy-efficient digital circuits, and systems, computer arithmetic, application-specific architecture design, and reconfigurable computing. He has published several journal and conference papers in these areas.